General Instructions for All Capstone Teams

1. Each team/group is expected to independently understand the assigned business domain and investigate the dataset provided.
2. The project objective will not be explicitly provided; teams are required to analyze the available data, identify potential business problems, and formulate suitable business and analytical objectives.
3. Teams must follow a structured data science lifecycle including business understanding, data understanding, data preparation, exploratory data analysis, problem discovery, feature engineering, model development, evaluation, and business recommendations.
4. Participants are encouraged to perform independent research using academic papers, industry reports, whitepapers, case studies, and other credible sources relevant to their assigned domain.
5. All assumptions made during the project must be clearly documented and justified.
6. Teams must maintain proper version control and documentation throughout the project lifecycle.
7. The use of Generative AI tools is permitted for learning and productivity purposes; however, all team members must thoroughly understand and be able to explain every aspect of the submitted work.
8. Teams must ensure that all code, analyses, visualizations, and recommendations are reproducible and well documented.
9. Any external datasets, libraries, models, frameworks, or reference materials used during the project must be appropriately acknowledged.
10. Teams are expected to focus not only on model performance but also on business relevance, interpretability, practical applicability, and potential business impact.

Pre-Submission Deliverables

Each team must submit:

- Capstone Pre-Submission Document (Maximum 3 Pages)
- Business Problem Understanding and Objective Formulation
- Challenges and Constraints
- Background Study / Literature Review
- Proposed Methodology / Solution Approach
- Workflow Diagram / Block Diagram
- Proposed Timeline
- Expected Deliverables and Outputs

Final Deliverables

Each team must submit the following:

- Complete Python Source Code
- Well-documented Jupyter Notebook(s) / python scripts
- Final Technical Report (Please follow a standard format)
- GitHub Repository Link

GitHub Submission Requirements

1. Every participant must create and maintain an individual GitHub repository.
2. Each participant must upload:
    - Source code
    - Notebooks
    - Reports / Presentation / Supporting documentation
3. The repository must contain a properly written README.md explaining:
    - Business problem
    - Dataset overview
    - Methodology
    - Results
    - Instructions for execution
4. Repository structure should be organized and professional.
5. Each participant must share their GitHub repository link before the final evaluation.

Presentation Requirements

- Every team member must participate in the final presentation.
- Every participant should be able to explain the business problem, methodology, model development process, results, and recommendations.
- Teams should be prepared for technical and business-oriented questions during evaluation.

Evaluation Expectations

Assessment will be based on:

- Business Understanding
- Problem Discovery
- Data Understanding and Quality Assessment
- Exploratory Data Analysis
- Methodology Design
- Feature Engineering
- Model Development
- Model Evaluation
- Business Impact
- Quality of Recommendations
- Documentation Quality
- GitHub Repository Quality
- Presentation and Communication Skills
- Individual Contribution

The objective of this capstone is not only to build a machine learning model, but to demonstrate the ability to solve a real-world business problem using data-driven approaches and communicate the findings effectively to stakeholders.